Class notes

Introduction to Functions

Relations, correlation, functions, domain, and range

Subject

Algebra I

Student

[redacted]

Teacher

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Today's lesson

Agenda

1. **Relations** — Tables, graphs, mapping diagrams
2. **Correlation & Variables** — Independent and dependent variables
3. **Identifying Functions** — The vertical line test
4. **Domain & Range** — Inequality and interval notation

Section 1

Relations

Ordered pairs, tables, graphs, and mapping diagrams

Notes

What Is a Relation?

Relation

A set of ordered pairs (x, y). Each pair links a value from the **domain** to a value from the **range**.

a coordinate (x, y)

a list with no repeats

all y vals

all x vals

For example: x is related to y in this way

x	1	2	3	4
y	5	6	7	8

→ (1,5) (2,6) (3,7) (4,8)

Notes

Domain and Range

When talking about functions, we like to talk about two properties of them: **domain** and **range**.

1. The **domain** is the set of all x values (input)

x	1	2	3	4
y	5	6	7	8

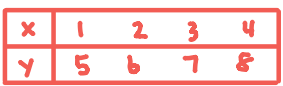
→ (1,5) (2,6) (3,7) (4,8)

2. The **range** is the set of all y values (output)

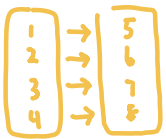
Notes

Three Ways to Show a Relation

Representation	How
① Table	write each x above its y
② Graph	plot each ordered pair as a point
③ Mapping diagram	arrow from each domain value to its range value







Example

Represent as a Table

Relation: $\{(-7, 1), (-3, 0), (-2, -1), (4, 7), (6, 4)\}$

domain →

range →

x	-7	-3	-2	4	6
y	1	0	-1	7	4

(-7,1) (-3,0) (-2,-1) (4,7) (6,4)

Question 1

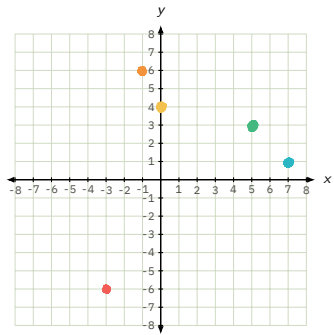
Table Practice

Relation: $\{(-1, -2),\ (0, -3),\ (0, 2),\ (6, -2),\ (7, -7)\}$

x	-1	0	0	6	7
y	-2	-3	2	-2	-7

Example

Represent as a Graph



Relation:

$\{(-3,-6), (-1,6), (0,4), (5,3), (7,1)\}$

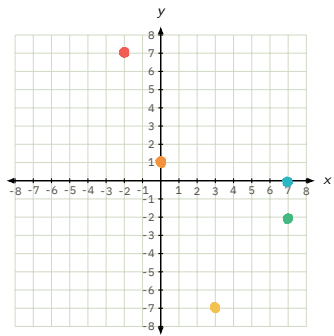
To find domain & range:

range → domain →

X	-3	-1	0	5	7
Y	-6	6	4	3	1

Question 1

Graph Practice



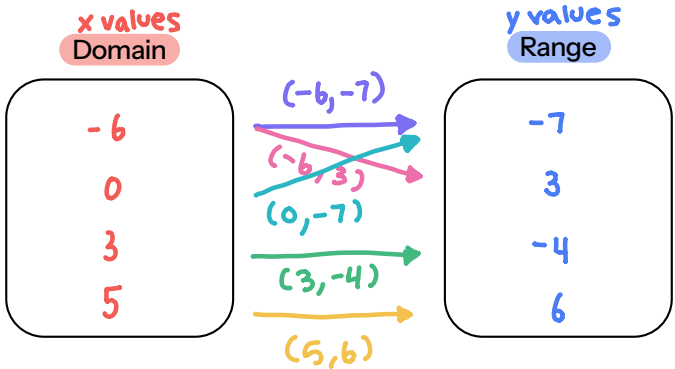
Relation:

$\{(-2, 7), (0, 1), (3, -7), (7, -2), (7, 0)\}$

Example

Represent as a Mapping Diagram

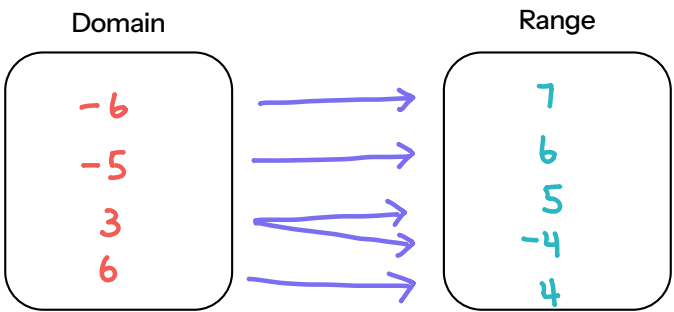
Relation: $\{(-6, -7),\ (-6, 3),\ (0, -7),\ (3, -4),\ (5, 6)\}$



Question 1

Mapping Diagram Practice

Relation: $\{(-6, 7),\ (-5, 6),\ (3, 5),\ (3, -4),\ (6, 4)\}$



Section 2

Correlation & Variables

How two quantities move together

Notes

Correlation & Variables

Independent & Dependent Variables

The **independent variable** changes on its own. The **dependent variable** responds to it.

it "depends" on the independent

「changes」 v/s 「is changed」

Notes

Correlation & Variables

Positive correlation	as one variable increases, the other increases
Negative correlation	as one variable increases, the other decreases
No correlation	the variables have 0 relationship

POSITIVE

as x↑, y↑



Age vs Reading Lvl.

- as age↑, lvl ↑
- as age ↓, lvl ↓

NEGATIVE

as x↑, y↓



Age vs Cavities

- as age↑, cavities ↓
- as age↓, cavities ↑

NONE

x, y share no relationship



Age vs Happiness

- people are happy and unhappy at all ages

Example

Speed & Stopping Distance

The speed of a car and the distance traveled to complete a stop after the brakes are applied.

Type of correlation	positive
Justification	as speed ↑, time to stop ↑ bc momentum
Independent variable	speed (thing that changes)
Dependent variable	breaking dist. (thing that is changed)



Question 1

Correlation Practice

The number of students present in class and the number of empty chairs.

Type of correlation	negative
Justification	as students ↑ empty seats ↓
Independent variable	students
Dependent variable	empty seats

n=1 student



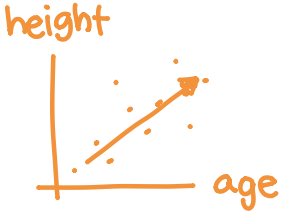
n=3 student



Question 2

Correlation Practice

Age (up to 12 years old) and height.



Type of correlation	positive
Justification	as age ↑ height ↑ (up to puberty)
Independent variable	age
Dependent variable	height

Question 3

Correlation Practice

The circumference of your head and your IQ.



Type of correlation	none
Justification	smart ppl have small heads too
Independent variable	head circumference
Dependent variable	IQ



Section 3

Identifying Functions

Ordered pairs, tables, and the vertical line test

Notes

What Is a Function?

Function

A relation where **every** input (domain value) is paired with **exactly one** output (range value).

NOT a FUNCTION



one input = multiple output

IS a FUNCTION



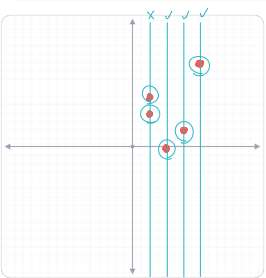
each input → 1 output

Notes

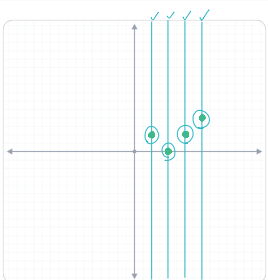
The Vertical Line Test

Vertical Line Test

If a vertical line crosses a graph more than once, the graph is **not** a function.



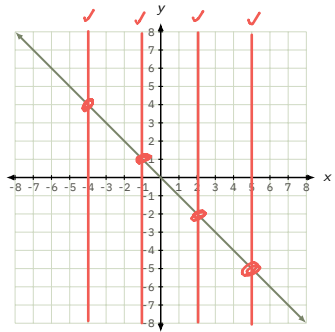
x	1	2	1	3	4
y	2	0	3	1	5



x	1	2	3	4
y	1	0	1	2

Example

Function or Not?



Is this relation a function?

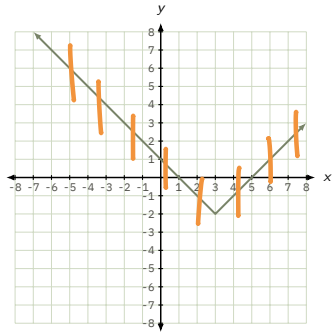
yes!

Why or why not?

passes v. line test

Question 1

Function or Not Practice



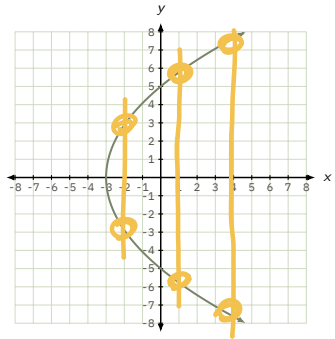
Is this relation a function?

yes!

Why or why not?

Question 2

Function or Not Practice



Is this relation a function?

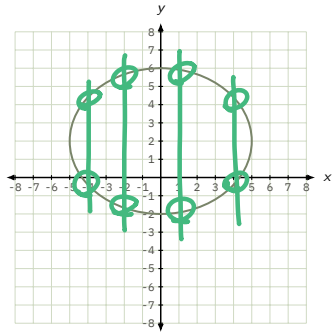
No!

Why or why not?

v. lines are crossed 2x

Question 3

Function or Not Practice



Is this relation a function?

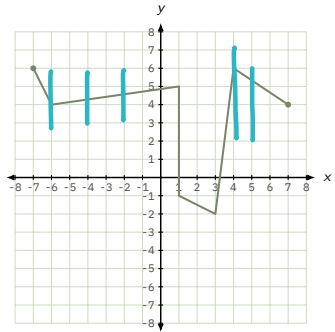
no!

Why or why not?

v lines cross 2x

Question 4

Function or Not Practice



Is this relation a function?

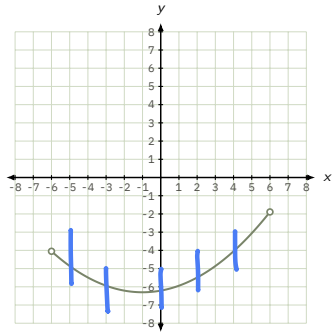
no!

Why or why not?

x=1 is a vertical line

Question 5

Function or Not Practice



Is this relation a function?

YES!

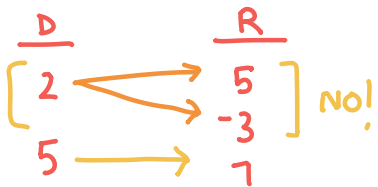
Why or why not?

only crosses

line 1x

Example

Domain, Range & Function



Relation: $\{(2,5),\ (2,-3),\ (5,7)\}$

Domain

$\{2, 5\}$

Range

$\{5, -3, 7\}$

Function?

NO! B/C (2,5) (2,-3)

Questions 1–3

Domain, Range & Function Practice

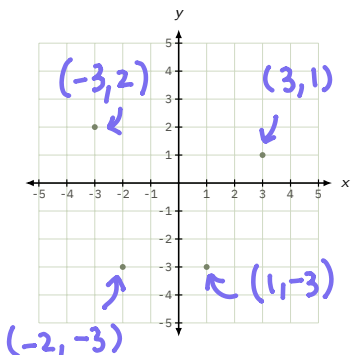
Relation	Domain	Range	Function?
$\{(8,5),\ (6,5),\ (7,2)\}$	$\{8,6,7\}$	$\{5,2\}$	✓
$\{(6,4),\ (8,-4),\ (9,10)\}$	$\{6,8,9\}$	$\{4,-4,10\}$	✓
$\{(9,9),\ (9,8),\ (9,7)\}$	$\{9\}$	$\{9,8,7\}$	✗



X

Example

Domain, Range & Function



Domain

{ −3, 3, 1, −2 }

Range

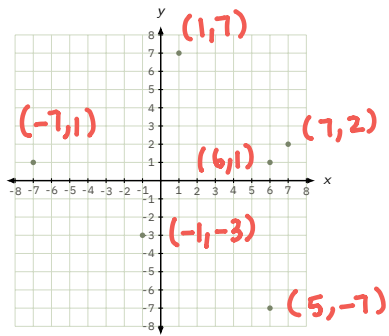
{ 2, 1, −3 }

Function?

yes!

Example

Read the Graph



Domain

{-7, 1, 6, 7, -1, 5}

Range

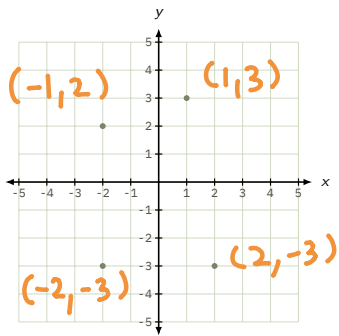
{1, 7, 2, -3, -7}

Function?

yes!

Question 1

Reading Graphs Practice



Domain

{ -1, 1, 2, -2 }

Range

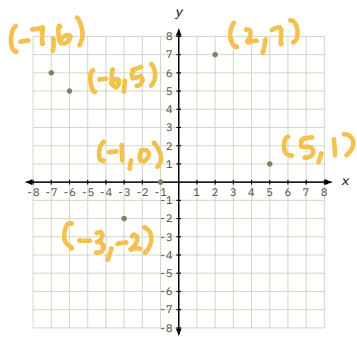
{ 2, 3, -3 }

Function?

yes!

Question 2

Reading Graphs Practice



Domain

{-7, -6, -1, 2, -3, 5}

Range

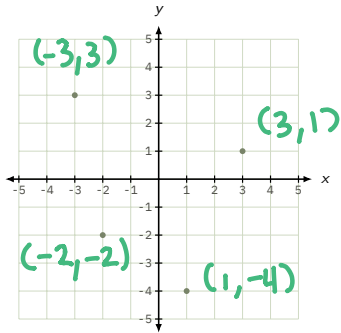
{6, 5, 0, 7, 1, -2}

Function?

yes!

Example

Range of the Function

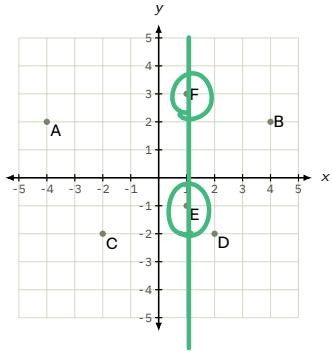


What is the range of the function shown?

$\{3, 1, -4, -2\}$

Example

Remove a Point to Make a Function



Which point, when deleted, makes the relation a function?

F OR E

Example

Passing Yards by Receiver

Receiver	Yards
Galindo	30
Hemphill	42
Harle	5
Carrillo	12
Segovia	10
Jenkins	12

Domain

here!

Range

{30, 42, 5, 12, 10}

Functional relationship?

yeah!

Question 1

Functional Relationship Practice

Height	Shoe size
$5'10''$	9
$5'11''$	9.5
$5'9''$	8.5
$6'1''$	12
$5'11''$	10
$6'0''$	11.5

Domain

//

Range

{8.5, 9, 9.5, 10, 11.5, 12}

Functional relationship?

NO!

Section 4

Domain & Range

Reading continuous graphs

Notes

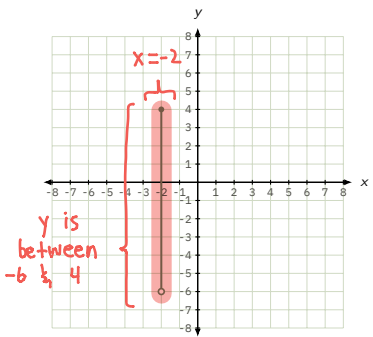
Domain & Range from a Graph

When a graph is continuous, write domain and range with **inequality notation**.

Domain	the x (input) values a graph covers
Range	the y (output) values a graph covers

Example

Function, Domain & Range



Function?

NO!

Domain

x = -2

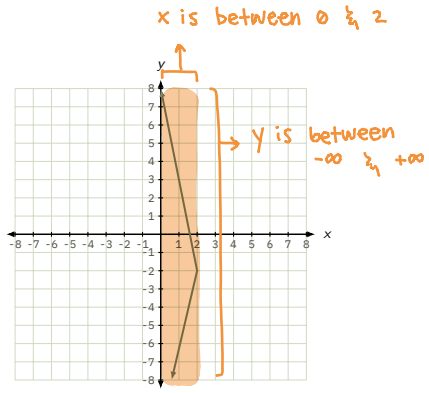
Range

can combine to $-6 < y \leq 4$

$y > -6$ $y \leq 4$

Question 1

Function, Domain & Range Practice



Function?

NO!

Domain

$x > 0$ $x \leq 2$

combine to $0 < x \leq 2$

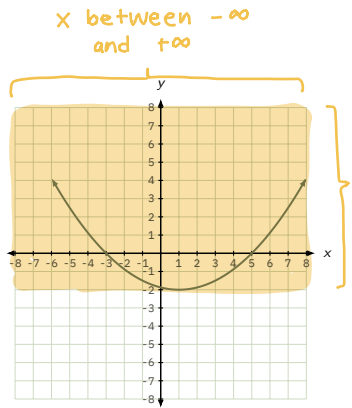
Range

$y > -\infty$ $y < +\infty$

combine to $-\infty < y < +\infty$

Question 2

Function, Domain & Range Practice



Function?

yes!

Domain

combine to $-\infty < x < +\infty$

$x > -\infty$ $\quad x < +\infty$

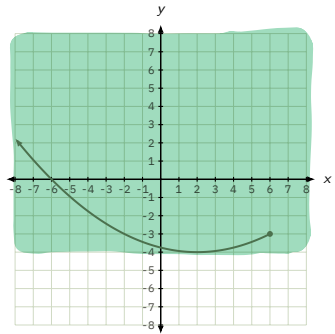
Range

combine to $-2 \leq y < +\infty$

$y \geq -2$ $\quad y < +\infty$

Question 3

Function, Domain & Range Practice



Function?

Yes!

Domain

min max

$-\infty < x \leq 6$

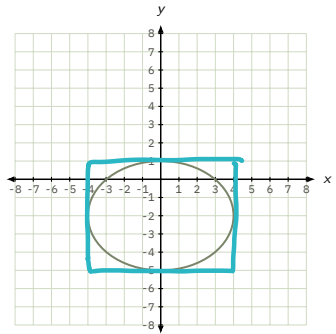
Range

min max

$-4 \leq y < +\infty$

Question 4

Function, Domain & Range Practice



Function?

no!

Domain

min max

$-4 \leq x \leq 4$

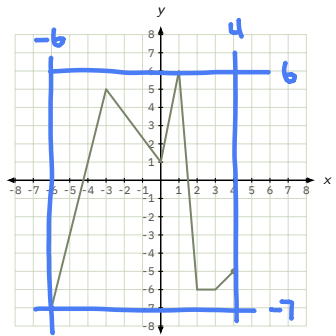
Range

min max

$-5 \leq y \leq 1$

Question 5

Function, Domain & Range Practice



Function?

Yes!

Domain

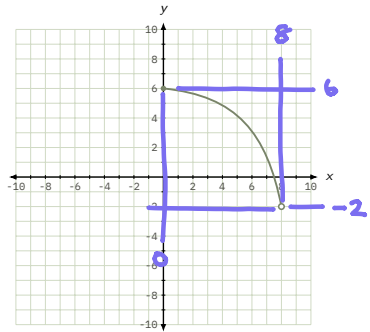
$-6 \leq x \leq 4$

Range

$-7 \leq y \leq 6$

Example

Domain & Range in Inequality Notation



Domain

$0 \leq x < 8$

Range

$-2 < y \leq 6$

Notes

Interval Notation

Symbol	Meaning	Matches
$[\ ]$	endpoint included	$\leq$, $\geq$, closed dot
$(\ )$	endpoint not included	$<$, $>$, open dot
$-\infty$, ∞	no end in that direction	always $(\ )$

$[$ min , max $)$

$3 < x < 5 \rightarrow (3, 5)$

$x \leq 10 \rightarrow (-\infty, 10]$

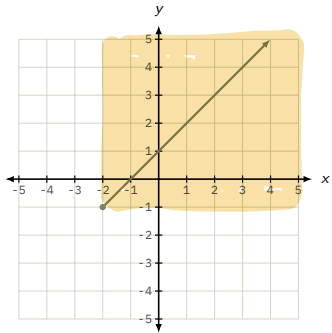
Notes

Inequality to Interval

Inequality	Interval
$-2 \leq x < 5$	$[-2, 5)$
$x > 3$	$(3, \infty)$
$x \leq -1$	$(-\infty, -1]$
all real numbers	$(-\infty, \infty)$

Example

Domain & Range in Interval Notation



Domain

$x \geq -2$

$[-2, +\infty)$

Range

$y \geq -1$

$[-1, +\infty)$

Questions 1–6

Interval Notation Practice

Write each in interval notation.

1. $1 \leq x \leq 7$

   min max

   [1 , 7]

2. $-4 < x < 0$

   min max

   (-4 , 0)

3. $x \geq -6$

   min

   [-6 , +∞)

4. $y < 2$

   max

   (-∞ , 2)

5. $-5 < y \leq 3$

   min max

   (-5 , 3]

6. all real numbers

   (-∞ , +∞)

Questions 7–10

Interval Notation Practice

Write each as an inequality in x.

7. $[0, 8)$

min max

$0 \leq x < 8$

8. $(-\infty, 4]$

min max

$-\infty < x \leq 4$

$x \leq 4$

9. $(-3, \infty)$

min max

$-3 < x < +\infty \rightarrow x > -3$

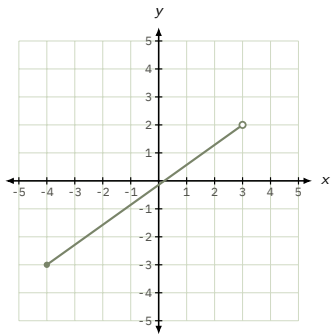
10. $[-1, 1]$

min max

$-1 \leq x \leq 1$

Question 11

Interval Notation Practice



Write the domain and range in interval notation.

Domain

min max

$-4 \leq x < 3$

$[-4, 3)$

Range

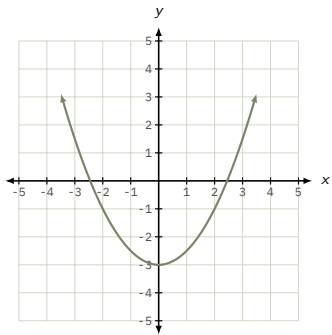
min max

$-3 \leq y < 2$

$[-3, 2)$

Question 12

Interval Notation Practice



Write the domain and range in interval notation.

Domain $-\infty$ to $+\infty$

$(-\infty, +\infty)$

Range -3 to $+\infty$

$[-3, +\infty)$